



Hotel Demand and Cancellation Forecast

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The Hotel Industry



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- Hotels have daily limited perishable inventory
 - Unsold rooms = Lost Revenue
 - Overselling rooms:
 - Pay for a comparable hotel
 - Transportation costs
 - Damage to reputation and loyalty



How Machine Learning Can Help

Metrics of Success:

- Maximize Revenue Per Available Room (RevPAR)
- Reaching a "perfect sell"

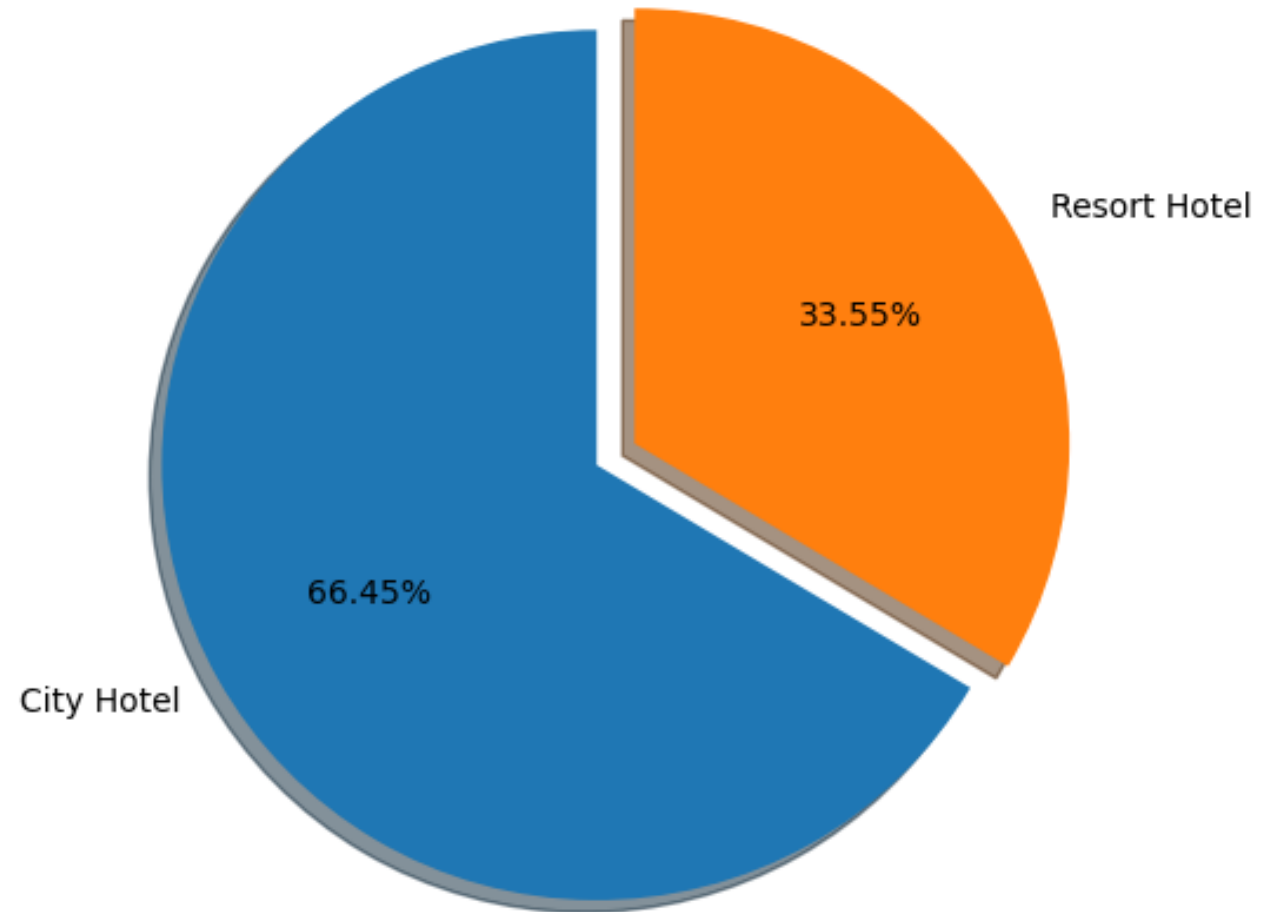
Predicting demand and cancellations to optimize pricing strategies

The Dataset

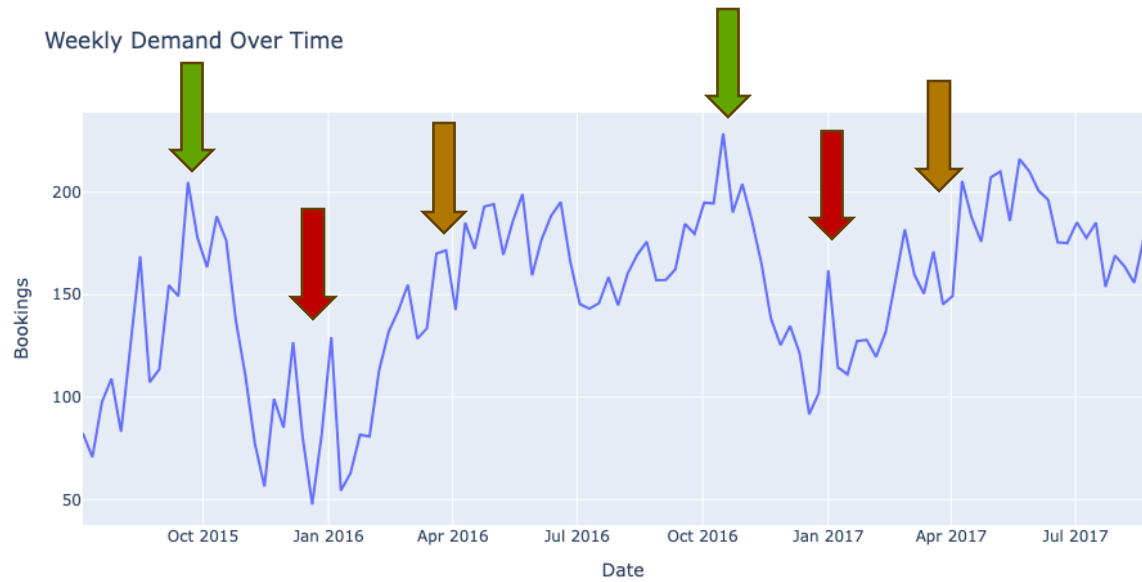
Important Columns:

- *is_canceled*
- *Arrival_date (year, month, day of month)*
- *Market_segment*
- *Lead_time*
- *Customer_type*

Percentage of Reservations by Hotel



Weekly Demand Over Time

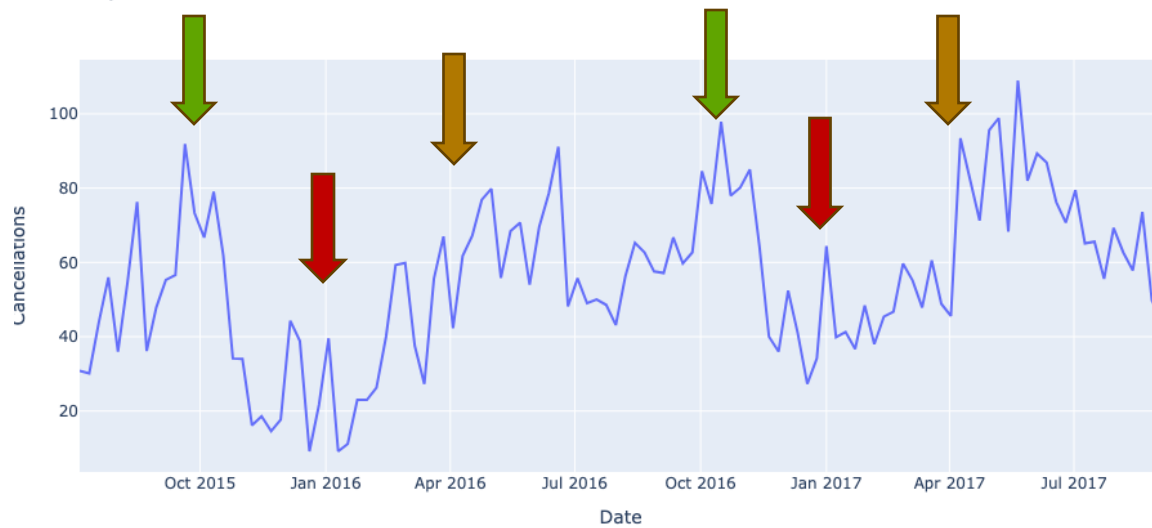


Initial EDA: Cancellation and Demand over Time

Seasonality seen between Demand
and Cancellations

- Peaks in October
- Valleys in January
- Increases in April

Weekly Cancellations Over Time



Model Comparisons



- Logistic Regression model outperformed Random Forest models in all metrics
- Low Precision:
 - Walking the guest + guest dissatisfaction
- Low Recall:
 - More unsold rooms
- Hotel A: Next closest hotel is far away
 - Prioritize **Precision** to prevent walking guests
- Hotel B: Centrally located in a big city
 - Prioritize **Recall** to maximize selling all rooms

Next Steps

- Explore baseline models 100% precision
- Further Modeling
 - XGBoost
 - Time Series Forecasting with Demand
- Hyperparameter Tuning



Thank You

